

DAX Aggregate Functions in Power BI

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Microsoft offers a service for business analytics called Power BI. It seeks to offer business intelligence capabilities and interactive visualizations with a user interface that is straightforward enough for users to build their own reports and dashboards. Here, we are using the dataset that contains information on the library supplies sold by a library manufacturer to different clients. All the relevant details like order date, client name, product name, product unit, etc. can be seen in it. Below is the link to download the dataset.

Dataset Used: [[SLS Order Details_Master](#), [SLS Order Details_Product Master](#), [Sheet1](#)]

DAX Aggregate Functions

It's possible for the mathematical operation to be a sum, average, maximum, count, or another. The process of combining values in your data is known as aggregating. An aggregate is the outcome of the mathematical procedure. Your data may be aggregated when Power BI service and Power BI Desktop build visuals.

In accordance with the expression, these functions compute a (scalar) value for each row in a column or table, such as count, sum, average, minimum, or maximum. Execution of the query creates a measure in the field. Drag a table from the visualizations and the measured field over it to get the output.

DAX Average

This function computes the average of the values contained in the provided column as input. A row is counted among the number of rows used as the divisor when a cell carries a value of 0, which is added to the sum of all the numbers. The row is not tallied when a cell is blank, though.

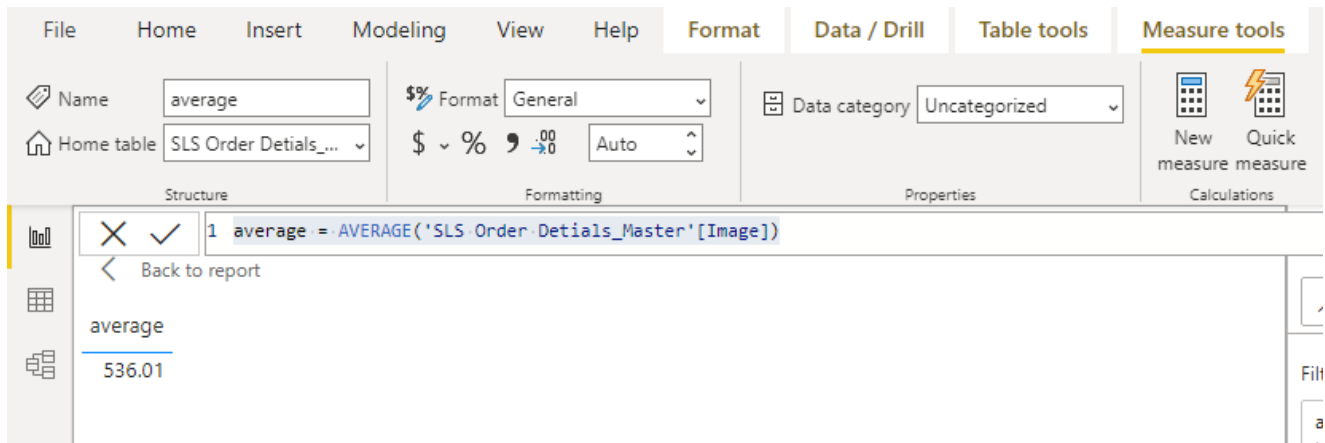
Syntax of DAX Average

| AVERAGE(<column>)

Example:

Here the function calculates the average number of orders received per day.

Formula: average = AVERAGE('SLS Order Detials_Master'[Image])



average

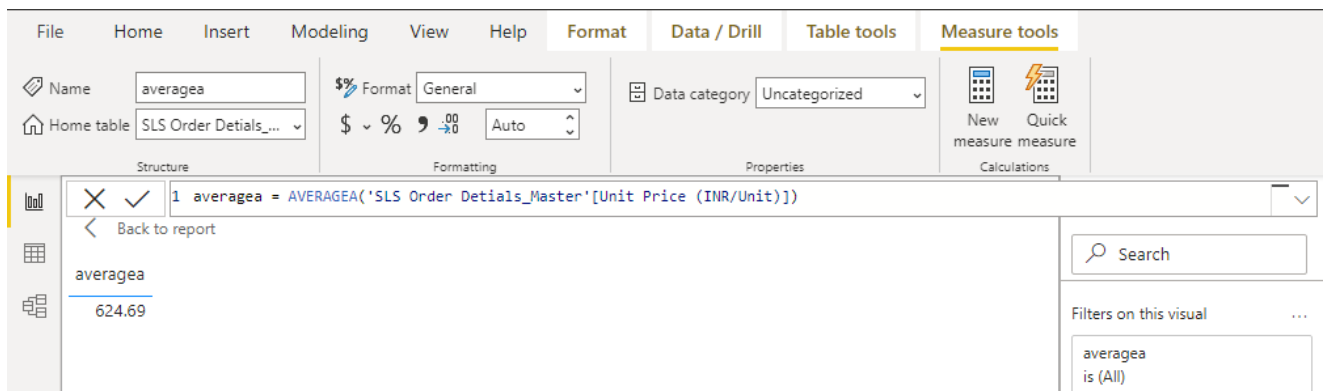
DAX AverageA

Gives the average (arithmetic mean) of the values in a column. The AVERAGEA function averages the values in a column and also manages non-numeric data types in accordance with the following guidelines:

- Values that are TRUE, count as 1.
- Empty text ("") counts as 0 (zero). Non-numeric text values have a count of 0 (zero).
- FALSE values are counted as 0 values (zero).

Syntax of DAX AverageA

AVERAGEA(<column>)



average a

Example:

Here the function calculates the average of the unit price of orders received per day.

Formula: `averagea = AVERAGEA('SLS Order Detials_Master'[Unit Price (INR/Unit)])`

DAX AverageX

Calculates the average (arithmetic mean) of a set of expressions evaluated over a table.

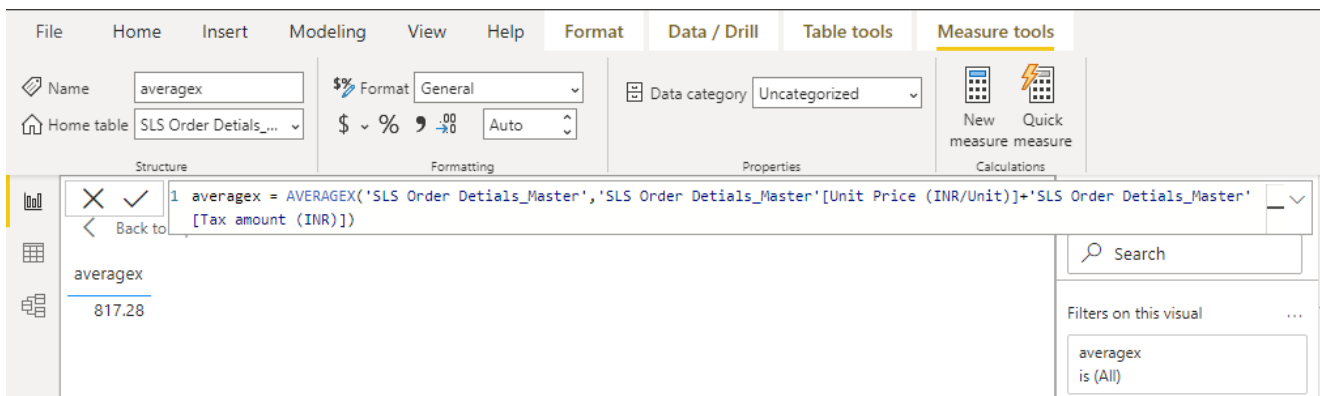
Syntax of DAX AverageX

| `AVERAGEX(<table>,<expression>)`

Example:

Evaluating the average of the column's value returned by adding the unit price of each product to the TAX incurred on it.

Formula: `averagex = AVERAGEX('SLS Order Detials_Master',
'SLS Order Detials_Master'[Unit Price (INR/Unit)]+
'SLS Order Detials_Master'[Tax amount (INR)])`



average x

DAX Count

This function only accepts a column as an argument. The following types of values are counted in rows using the COUNT function:

- Numbers
- Dates
- Strings

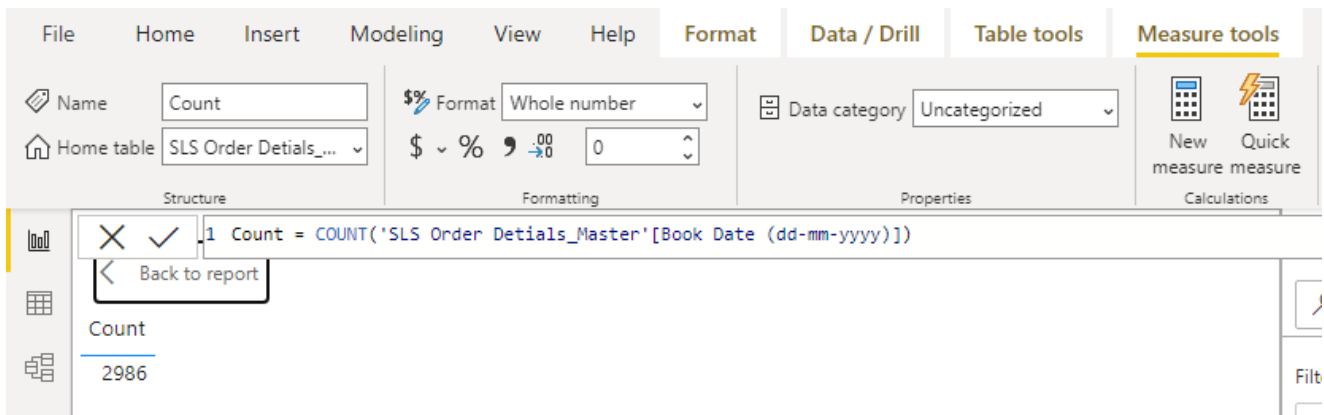
Syntax of DAX Count

| `COUNT(<column>)`

Example:

Counting the number of orders as per date.

Formula: Count = COUNT('SLS Order Detials_Master' [Book Date (dd-mm-yyyy)])



count

DAX CountA

Determines how many rows in the chosen column have non-blank values. The function returns a blank if it cannot locate any rows to count.

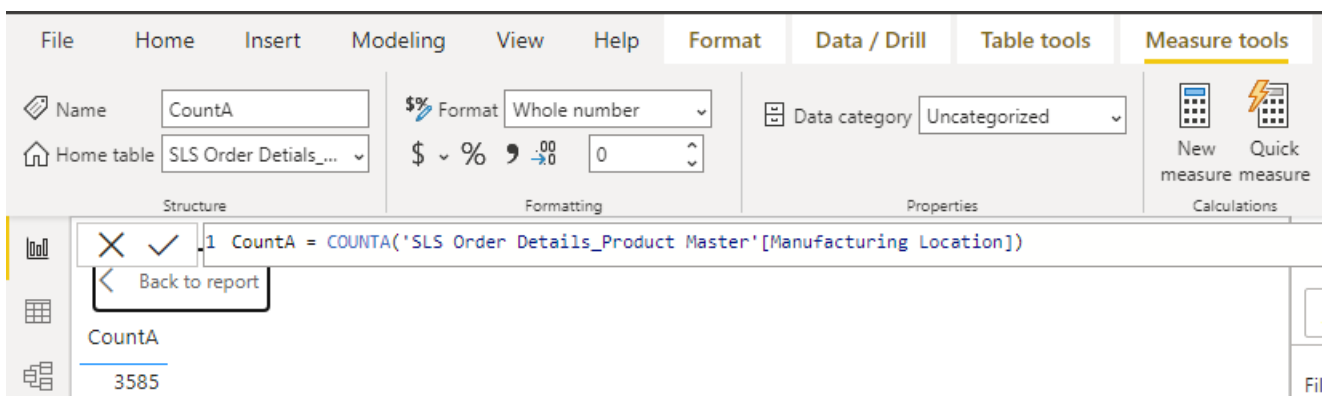
Syntax of DAX CountA

COUNTA(<column>)

Example:

Counting the number of products that have manufacturing locations specified on them.

Formula: CountA = COUNTA('SLS Order Detials_Master' [Manufacturing Location])



countA

DAX CountX

When evaluating an expression over a table, counts the number of rows that contain a number or an expression that evaluates to a number.

It can be used to calculate the counts of rows based on certain expressions within FILTER.

Two arguments are needed for the COUNTX function. A table or any expression returning a table must always be the first argument. The column or expression that COUNTX searches for is the second input

Syntax of DAX CountX

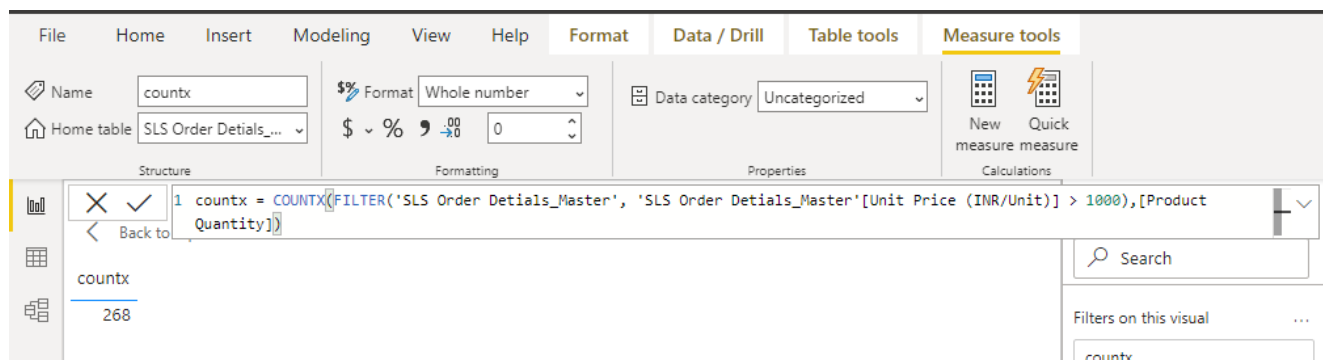
```
COUNTX(<table, expression>)
```

```
COUNTX(FILTER(<table, expression>, [column of which counts needs to be  
returned]))
```

Example:

Returning the number of products that have a unit price greater than 1000.

Formula: CountX = COUNTX(FILTER('SLS Order Detials_Master',
'SLS Order Detials_Master'[Unit Price (INR/Unit)] > 1000),
[Product Quantity]))



countX

DAX CountAX

Counts non-blank results when evaluating the result of an expression over a table.

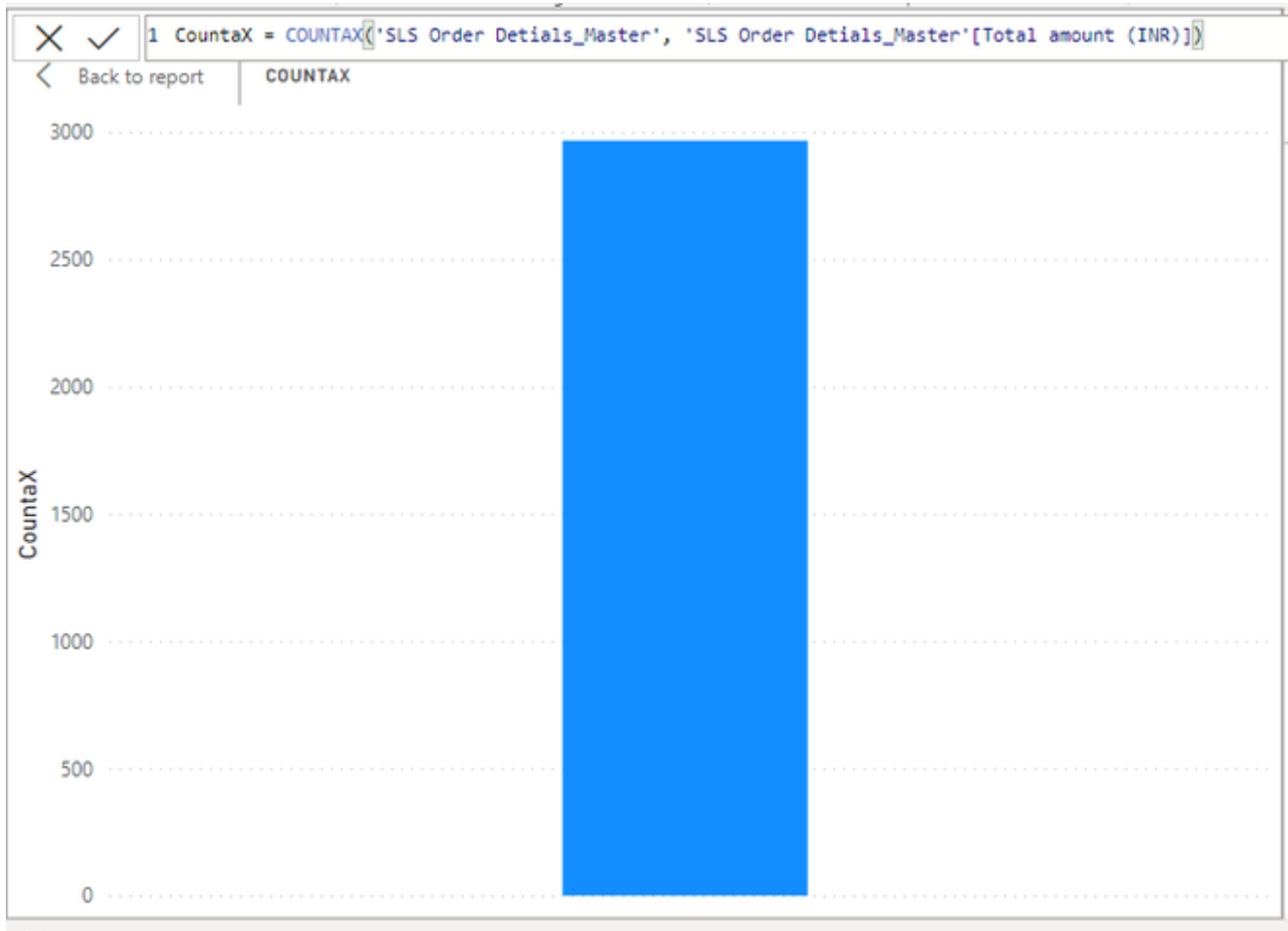
When determining the outcome of an expression over a table, the COUNTAX function counts results that are not blank. In other words, it functions just like the COUNTA function but is used to loop across table rows and count the rows where the supplied expressions return a result that is not blank.

Syntax of DAX CountAX

```
COUNTAX(<table, expression>)
```

Example:

Formula: CountAX = COUNTAX('SLS Order Detials_Master', 'SLS Order Detials_Master'[Total amount (INR)])



DAX CountBlank

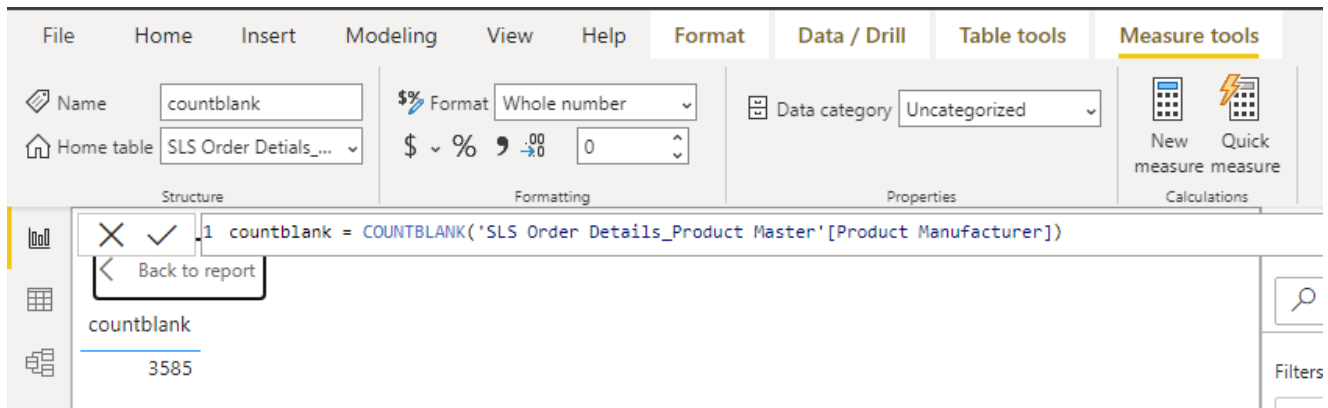
A column is the only argument this function accepts. Columns can include any kind of data, but only blank cells are counted. Due to the fact that zero is a valid numeric value and not a blank cell, cells with the value zero (0) are not counted.

Syntax of DAX CountBlank

| COUNTBLANK(<column>)

Example:

Formula: countblank = COUNTBLANK('SLS Order Detials_Master'[Product Manufacturer])



count blank

DAX CountRows

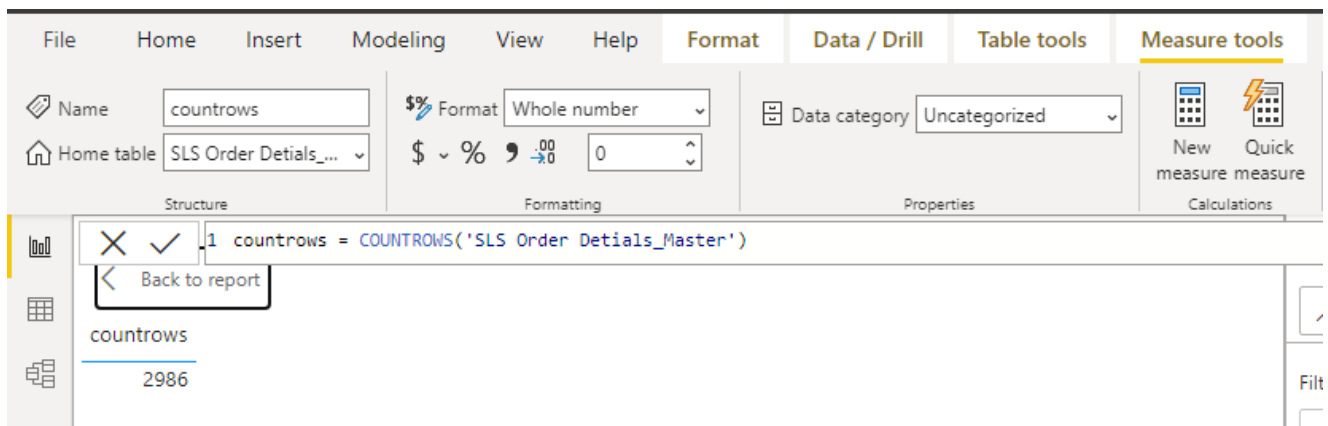
Determines how many rows there are in the supplied table or a table that has been defined using an expression.

Syntax of DAX CountRows

COUNTROWS([<table>])

Example:

Formula: COUNTROWS('SLS Order Details_Master')



count rows

DAX DistinctCount

Determines how many unique values there are in a column.

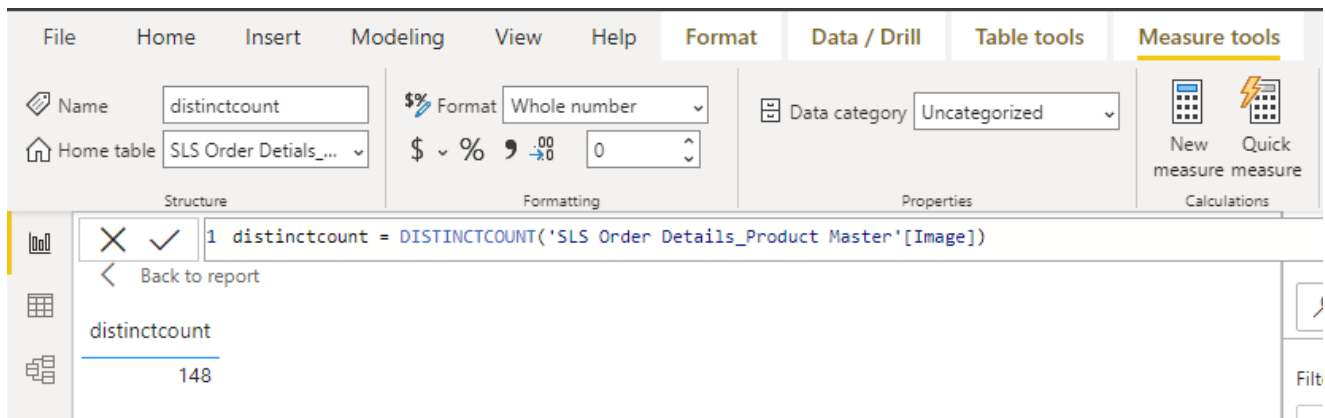
This function only accepts a column as an argument. You can utilize columns with any kind of data in them. The function returns a BLANK if there are no rows to count; otherwise, it gives the number of distinct values.

Syntax of DAX DistinctCount

| DISTINCTCOUNT(<column>)

Example:

Formula: distinctcount = DISTINCTCOUNT('SLS Order Detials_Product Master'[Image])



distinct count

DAX DistinctCountNoBlank

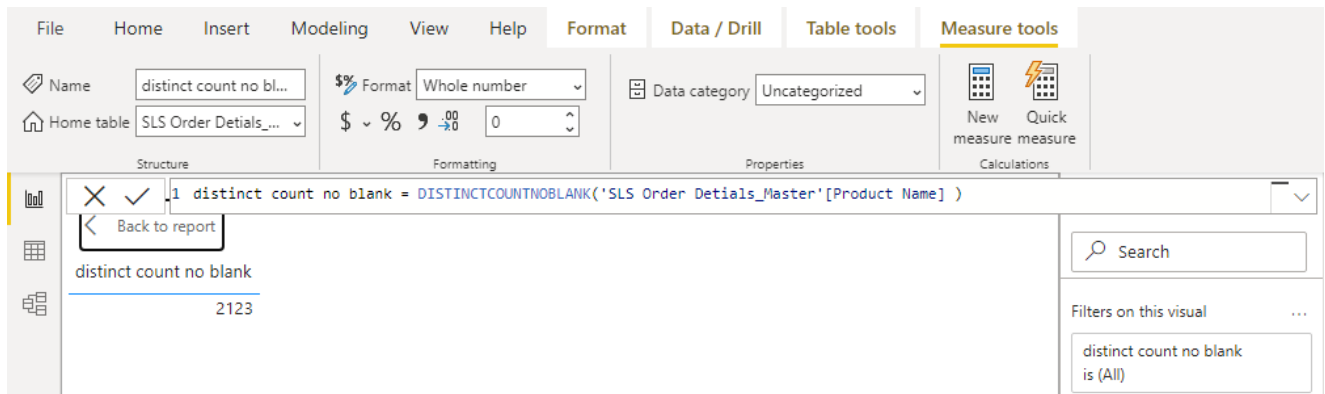
Counts the number of distinct values in a column.

Syntax of DAX DistinctCountNoBlank

| DISTICTCOUNTNOBLANK(<column>)

Example:

Formula: distinct count no blank = DISTINCTCOUNTNOBLANK('SLS Order Detials_Master'[Product Name])



distinct count no blank

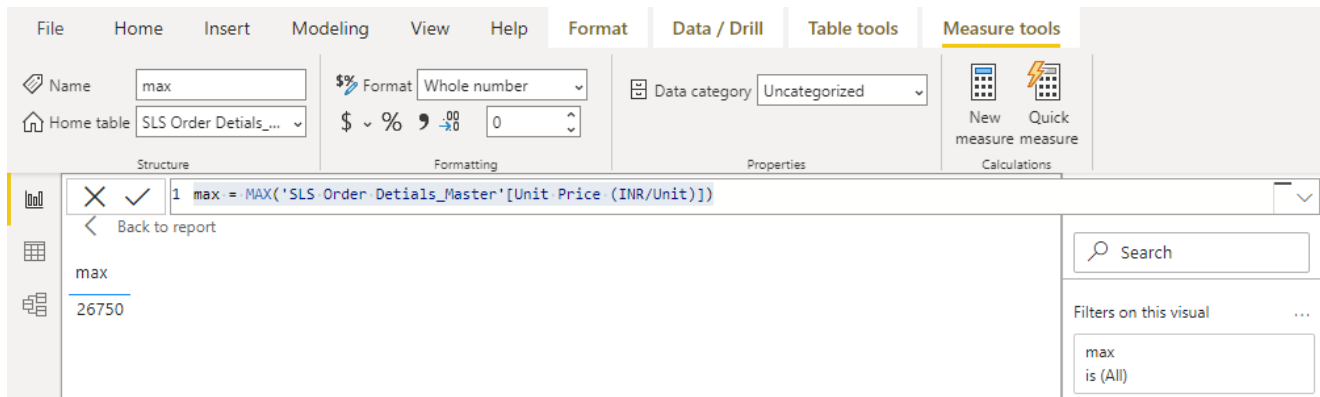
DAX Max

Returns the largest numeric value in a column, or between two scalar expressions.

Syntax of DAX Max

MAX(<column>) /

MAX(<expression1, expression2>)



max

Example:

Formula: max = MAX('SLS Order Detials_Master'[Unit Price (INR/Unit)])

DAX MaxA

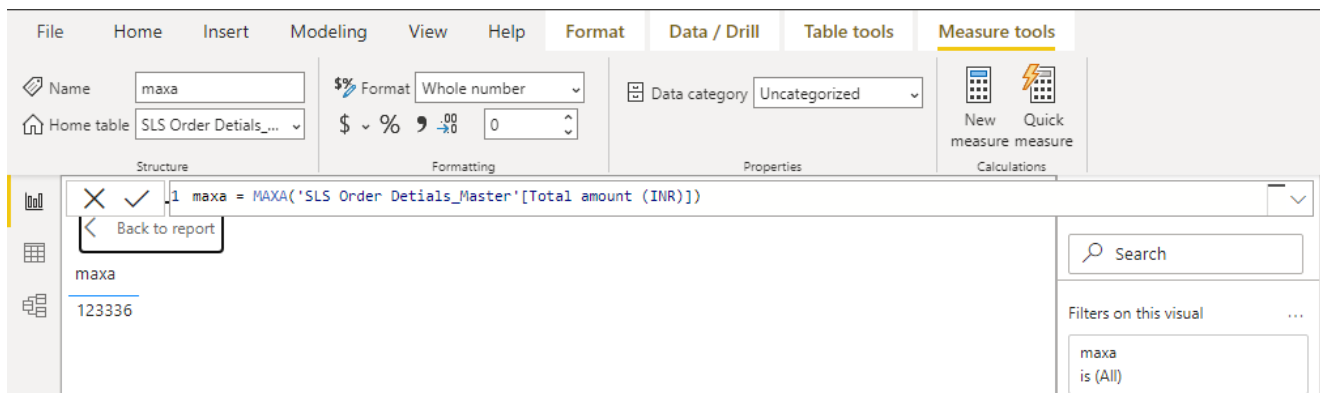
Returns the largest value in a column.

Syntax of DAX MaxA

MAXA(<column>)

Example:

Formula: maxa = MAXA('SLS Order Detials_Master'[Total amount (INR)])



max a

DAX Maxx

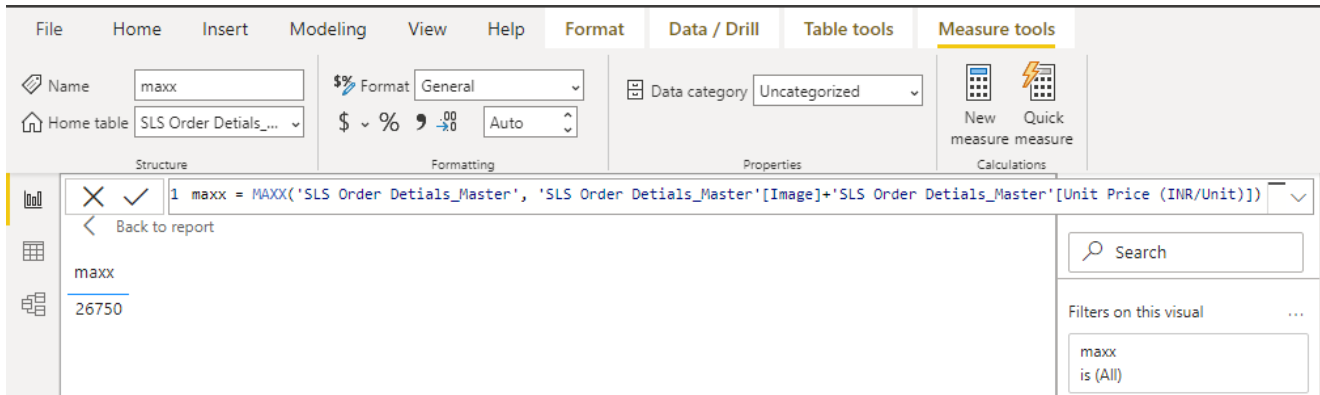
Evaluates an expression for each row of a table and return the largest numeric value.

Syntax of DAX Maxx

MAXX(<table, expression>)

Example:

Formula: maxx = MAXX('SLS Order Detials_Master',
'SLS Order Detials_Master'[Image] +
'SLS Order Detials_Master'[Unit Price (INR/Unit)])



max x

DAX Min

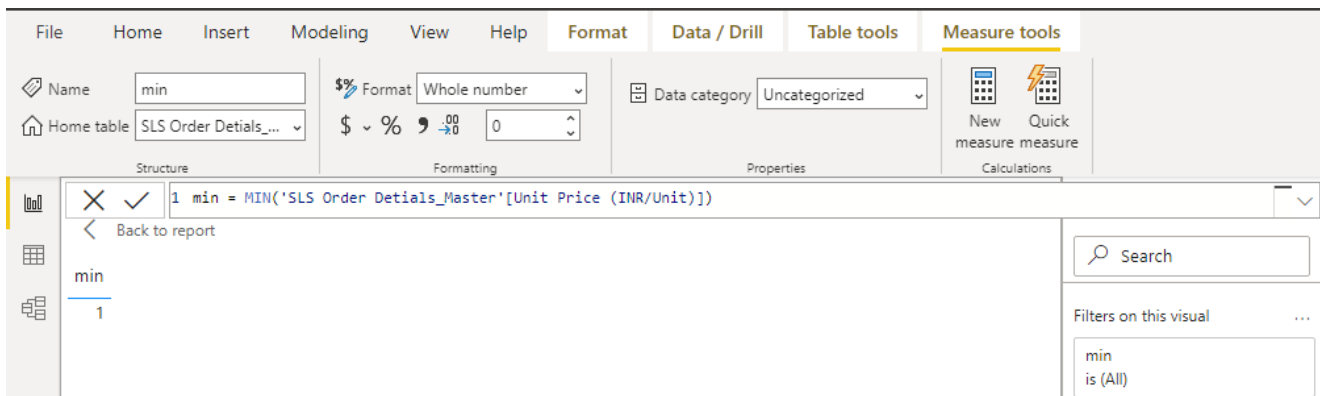
Returns the smallest numeric value in a column, or between two scalar expressions.

Syntax of DAX Min

MIN(<column>)

Example:

Formula: min = MIN('SLS Order Detials_Master'[Unit Price (INR/Unit)])



min

DAX MinA

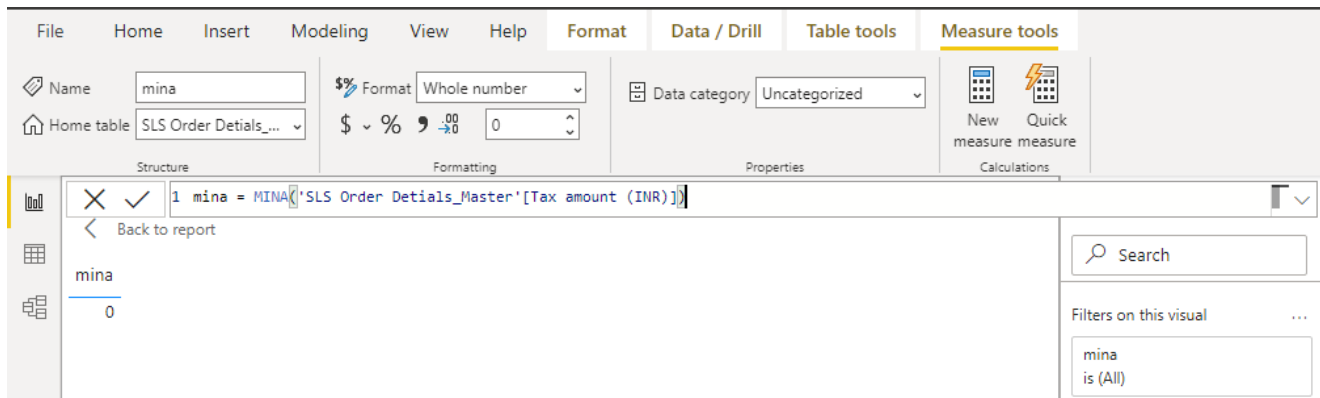
Returns the smallest value in a column, including any logical values and numbers represented as text.

Syntax of DAX MinA

MINA(<column>)

Example:

Formula: mina = MINA('SLS Order Detials_Master'[Tax amount (INR)])



min a

DAX MinX

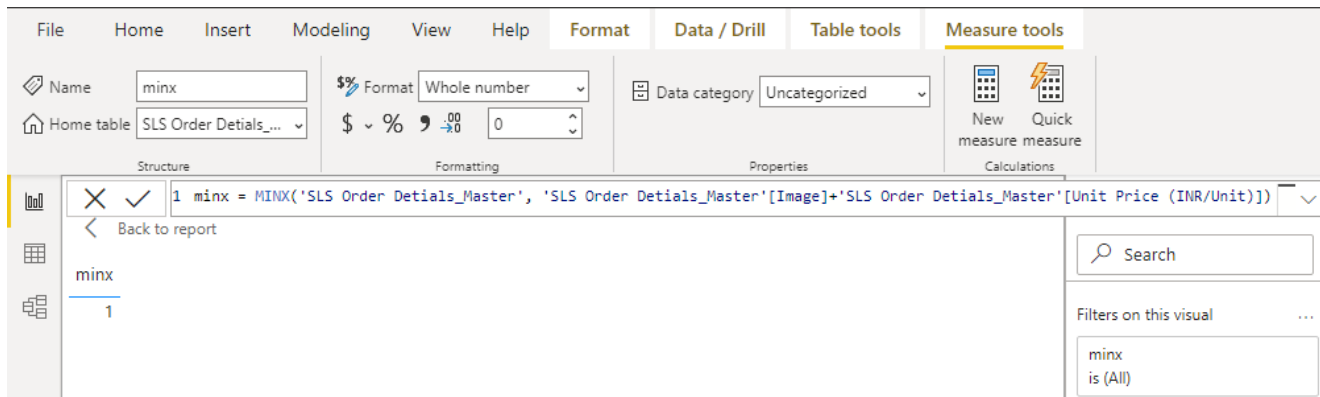
Returns the smallest numeric value that results from evaluating an expression for each row of a table.

Syntax of DAX MinX

MINX(<table, expression>)

Example:

Formula: minx = MINX('SLS Order Detials_Master',
'SLS Order Detials_Master'[Image] +
'SLS Order Detials_Master'[Unit Price (INR/Unit)])



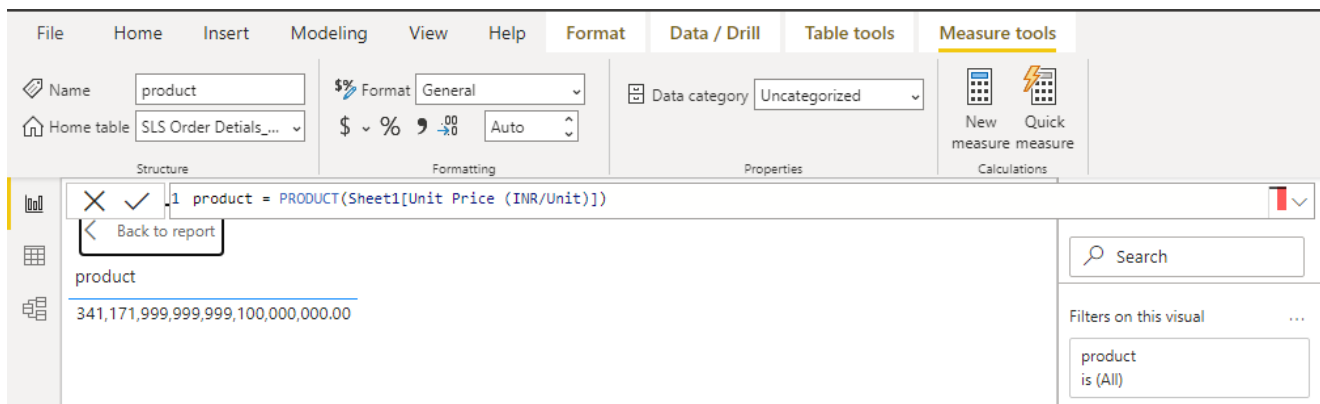
min x

DAX Product

Returns the product of the numbers in a column.

Syntax of DAX Product

| `PRODUCT(<column>)`



product

Example:

Formula: `product = PRODUCT(Sheet1[Unit Price (INR/Unit)])`

DAX ProductX

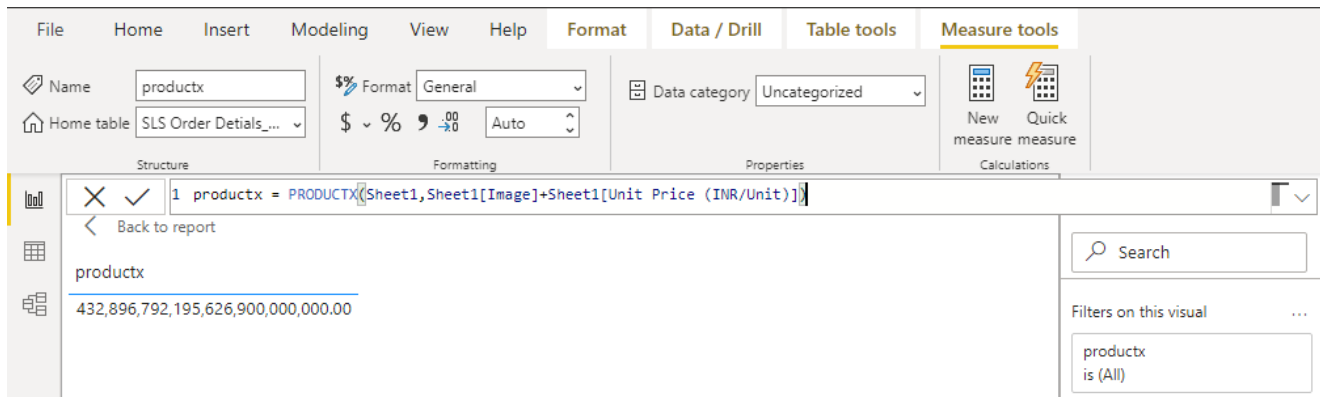
Returns the product of an expression evaluated for each row in a table.

Syntax of DAX ProductX

| `PRODUCTX(<table, expression>)`

Example:

Formula: `productx = PRODUCTX(Sheet1, Sheet1[Image]+Sheet1[Unit Price (INR/Unit)])`



product x

DAX Sum

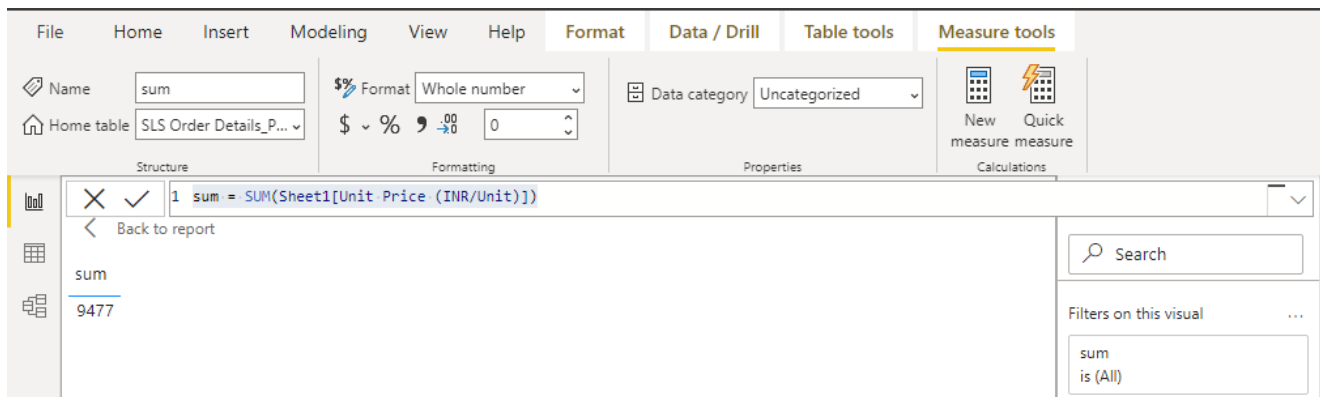
Adds all the numbers in a column.

Syntax of DAX Sum

| SUM(<column>)

Example:

Formula: sum = SUM(Sheet1[Unit Price (INR/Unit)])



sum

DAX SumX

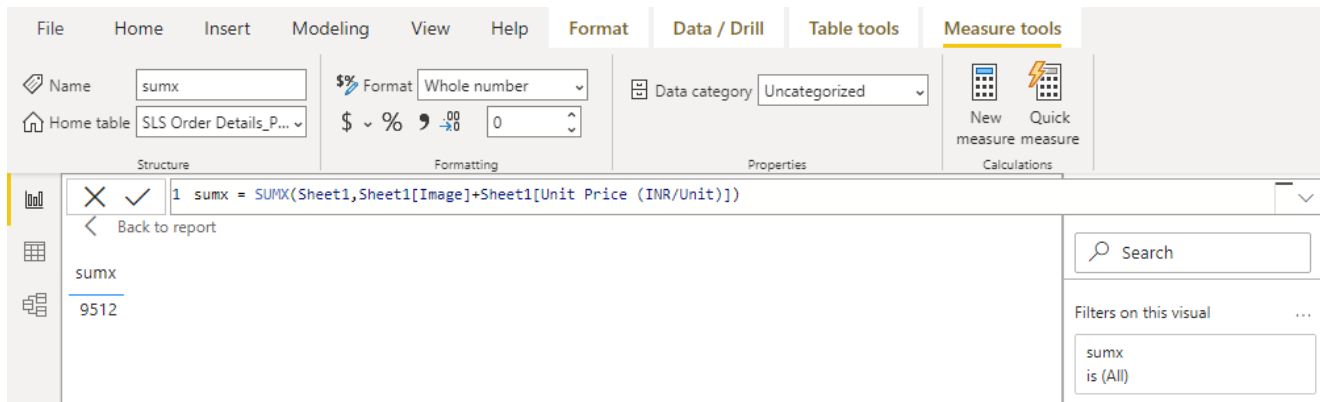
Returns the sum of an expression evaluated for each row in a table.

Syntax of DAX SumX

| SUMX(<table, expression>)

Example:

Formula: sumx = SUMX(Sheet1,Sheet1[Image]+Sheet1[Unit Price (INR/Unit)])



sum x

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